

Shaswata Das

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Go-first Systems Engineer specializing in high-throughput, stateless service architecture and production-grade rewrites of complex legacy systems. Sole engineer behind a ground-up Go rewrite of a production Identity Provider serving enterprise financial clients — reducing container footprint from GB-scale IIS to a 44 MB static binary and enabling horizontal autoscaling on Kubernetes. Proven track record in distributed systems, security-critical auth flows (OAuth 2.0, SAML, OIDC, MFA), and AI-driven platforms across the fintech and tax-compliance domain. Seeking high-impact systems and platform engineering roles at globally distributed, high-performance technology companies.

TECHNICAL SKILLS

Languages & Frameworks: Go (Golang / Chi), Python, C# (.NET 4.7.2 – 8/9), TypeScript, Next.js.

Backend Systems: Stateless Microservices, CQRS, Event-Driven Architecture, REST/gRPC, Low-Latency Design, Redis.

Security & Identity: OAuth 2.0, SAML (SP + IdP), OpenID Connect, SCIM, RFC 6238 TOTP/MFA, OWASP, RBAC.

AI & Infrastructure: LangChain, RAG, Vector DBs, Docker, Kubernetes (k8s), Azure, AWS, CI/CD Pipelines.

PROFESSIONAL EXPERIENCE

Software Engineer

Orbitax Bangladesh Limited | Mar 2024 – Present

- **Production Go Rewrite — Identity Provider (Solo):** Sole engineer on a ground-up rewrite of the production Identity Provider from **ASP.NET MVC 5 / .NET 4.7.2** (527 files, 32k LOC, 63 controllers, IIS-hosted) to **Go + Chi** (304 files, 43.5k LOC). Delivered full feature parity — **OAuth 2.0, OIDC, SAML (SP + IdP), SCIM, MFA/TOTP, magic-link, device registration, password reset** — plus a bundled Next.js frontend. Measurable outcomes: dependency surface cut **5×** (117 NuGet → 23 Go modules); artifact reduced from 82 DLLs / 32.8 MB to a **single 44 MB static binary**; container base from Windows+IIS (GB-scale) to **distroless/static ≈2 MB**; process start **0.12 s**; eliminated SQL Server entirely (Mongo-only); replaced commercial SAML license (ComponentSpace) with MIT **crewjam/saml**. Stateless design enables **horizontal autoscaling** (2–6 pods) on Kubernetes where the legacy was session-affinity-pinned to a single VM. Completed in **~8 weeks** against a 15–16 week estimate, 161 commits.
- **Technical Leadership & Mentorship:** Provided technical direction and code-review standards for junior engineers across the Identity and AdminPanel surfaces; sole production owner of the IdP, including incident response.
- **Cross-Platform Microservices (Python CQRS):** Engineered a **CQRS-based microservice in Python** with full protocol compatibility with existing .NET systems. Built a custom library bridging **SignalR** real-time messaging with **RabbitMQ**, enabling bidirectional polyglot communication. Integrated **Redis** for distributed caching.
- **AI Agent Engineering (XatBot):** Architected the backend for "XatBot," a production AI tax assistant using **LLMs** and **RAG** over 70k+ legal documents. Developed usage reporting modules for agent performance and engagement metrics. (*Stack: C#, Python, OpenAI, Azure, Qdrant*)
- **Database Modernization & Cost Optimization:** Led migration of the core Identity database from SQL Server + Entity Framework to **MongoDB**, accelerating feature cycles by 40%. Developed automated backup/restore workflows for database, S3, and persistent storage, cutting infrastructure costs for inactive workspaces by 35%.

Associate Software Engineer

Orbitax Bangladesh Limited | Mar 2022 – Feb 2024

- **Full Ownership — Legacy Identity Provider:** Became sole owner of the production IdP (enterprise financial/tax-compliance clients) after remediating critical **SAML SSO** misconfigurations surfaced by a **Big 4 audit**. Stabilised the "MagicLink" passwordless login system and implemented **RFC 6238 TOTP MFA**, raising the security baseline across all enterprise accounts.
- **Infrastructure Hardening:** Decoupled the Administrative Application onto an isolated domain, reducing the public attack surface. Designed secure token validation scopes and enforced granular **RBAC** permission scopes.
- **Feature Engineering (Financial Platform):** Engineered the Net Operating Loss calculation module and rating system for the ITP Desktop client (**Angular + C# .NET**), directly impacting revenue-facing user retention on a financial compliance product used by global enterprises.

Software Engineer Intern (Converted to Full Time)

Orbitax Bangladesh Limited | Sep 2021 – Feb 2022

- **Go Performance Engineering (R&D Port):** Ported a high-throughput .NET microservice to **Go (Golang)**, achieving a **60% reduction in latency** and **70% reduction in memory usage** — establishing Go as a viable production language at the company and laying the groundwork for later full-system rewrites.
- **Internal Tools & QA Automation:** Built the "Open Data Extractor Web" node for the internal Executable Actions Tool (EAT) and created batch scripting infrastructure for automated **JMeter** load tests, streamlining release validation.

PROJECTS & RESEARCH CONTRIBUTIONS

Cirq (Google Quantum AI) – Contributor

2024

- Clarified documentation and parameterization of **CZPowGate**, merged into the main repository. **PR Link**

microOS (Systems Engineering)

GitHub

- Minimal OS with loader, bootstrap, IDT, and kernel. Runs on QEMU/bare metal; demonstrates systems engineering fundamentals. **Link**

XatBot (Orbitax)

Production

- AI-powered tax assistant grounded in 70k+ legal documents; integrates retrieval-augmented generation for domain-specific accuracy. **Link**

EDUCATION

Bachelor of Science in Software Engineering

Feb 2023

Shahjalal University of Science & Technology

Sylhet, Bangladesh